



4.1 THE WORKING PRINCIPLE OF A TRANSISTOR

MIKROELEKTRONIKA

Transistors are used in analog circuits to amplify a signal. They are also used in power supplies as a regulator and you will also find them used as a switch in digital circuits.

The best way to explore the basics of transistors is by experimenting. A simple circuit is shown below. It uses a power transistor to illuminate a globe. You will also need a battery, a small light bulb (taken from a flashlight) with properties near 4.5V/0.3A, a linear potentiometer (5k) and a 470 ohm resistor. These components should be connected as shown in figure 4.4a.

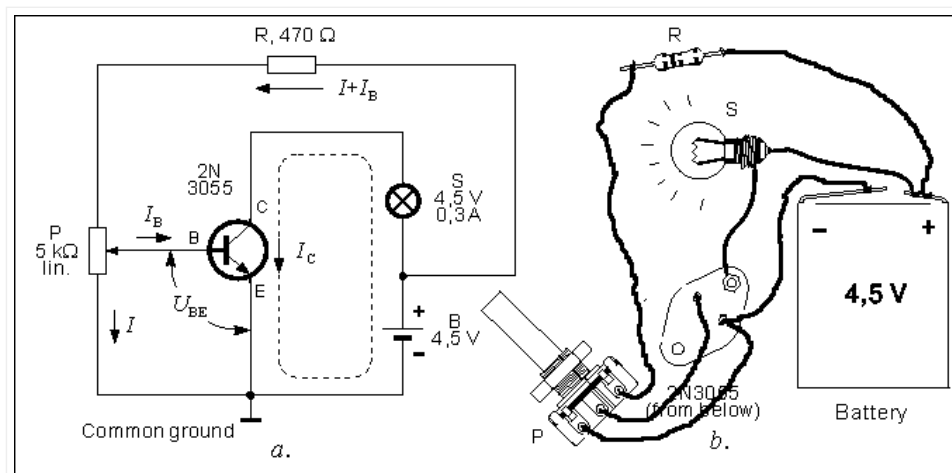


Fig. 4.4: Working principle of a transistor: potentiometer moves toward its upper position – voltage on the base increases – current through the base increases – current through the collector increases – the brightness of the globe increases.

Resistor (R) isn't really necessary, but if you don't use it, you mustn't turn the potentiometer (pot) to its high position, because that would destroy the transistor – this is because the DC voltage U_{BE} (voltage between the base and the emitter), should not be higher than 0.6V, for silicon transistors.

Turn the potentiometer to its lowest position. This brings the voltage on the base (or more correctly between the base and ground) to zero volts ($U_{BE} = 0$). The bulb doesn't light, which means there is no current passing through the transistor.

As we already mentioned, the potentiometers lowest position means that U_{BE} is equal to zero. When we turn the knob from its lowest position U_{BE} gradually increases. When U_{BE} reaches 0.6v, current starts to enter the transistor and the globe starts to glow. As the pot is turned further, the voltage on the base remains at 0.6v but the current increases and this increases the current through the collector-emitter circuit. If the pot is turned fully, the base voltage will increase slightly to about 0.75v but the current will increase significantly and the globe will glow brightly.

If we connected an ammeter between the collector and the bulb (to measure I_C), another ammeter between the pot and the base (for measuring I_B), and a voltmeter between the ground and the base and repeat the whole experiment, we will find some interesting data. When the pot is in its low position U_{BE} is equal to 0V, as well as currents I_C and I_B . When the pot is turned, these values start to rise until the bulb starts to glow when they are: $U_{BE} = 0.6V$, $I_B = 0.8mA$ and $I_C = 36 mA$ (if your values differ from these values, it is because the 2N3055 the writer used doesn't have the same specifications as the one you use, which is common when working with transistors).

The end result we get from this experiment is that when the current on the base is changed, current on the collector is changed as well.

Let's look at another experiment which will broaden our knowledge of the transistor. It requires a BC107 transistor (or any similar low power transistor), supply source (same as in previous experiment), 1M resistor, headphones and an electrolytic capacitor whose value may range between 10u to 100μF with any operating voltage. A simple low frequency amplifier can be built from these components as shown in diagram 4.5.

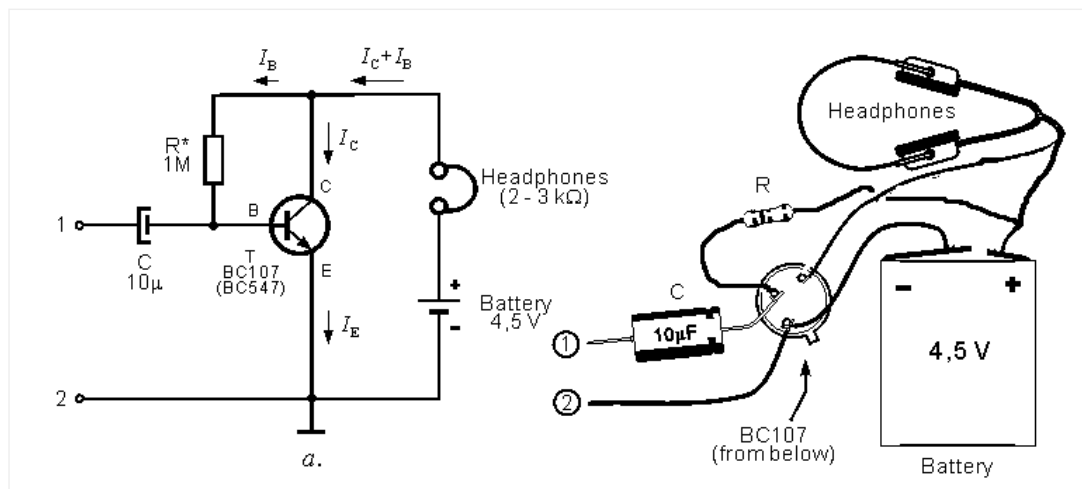


Fig. 4.5: A simple transistor amplifier

It should be noted that the schematic 4.5a is similar to the one on 4.4a. The main difference is that the collector is connected to headphones. The “turn-on” resistor – the resistor on the base, is 1M. When there is no resistor, there is no current flow I_B , and no I_C current. When the resistor is connected to the circuit, base voltage is equal to 0.6V, and the base current $I_B = 4\mu A$. The transistor has a gain of 250 and this means the collector current will be 1 mA. Since both of these currents enter the transistor, it is obvious that the emitter current is equal to $I_E = I_C + I_B$. And since the base current is in most cases insignificant compared to the collector current, it is considered that:

$$I_C = I_E.$$

The relationship between the current flowing through the collector and the current flowing through the base is called the transistor's current amplification coefficient, and is marked as h_{FE} . In our example, this coefficient is equal to:

$$h_{FE} = \frac{I_C}{I_B} = \frac{1mA}{4\mu A} = \frac{1 \cdot 10^{-3} A}{4 \cdot 10^{-6} A} = 250.$$

Put the headphones on and place a fingertip on point 1. You will hear a noise. Your body picks up the 50Hz AC “mains” voltage. The noise heard from the headphones is that voltage, only amplified by the transistor. Let's explain this circuit a bit more. AC voltage with frequency 50Hz is connected to transistor's base via the capacitor C. Voltage on the base is now equal to the sum of a DC voltage (0.6 approx.) via resistor R, and AC voltage “from” the finger. This means that this base voltage is higher than 0.6V, fifty times per second, and fifty times

slightly lower than that. Because of this, current on the collector is higher than 1mA fifty times per second, and fifty times lower. This variable current is used to shift the membrane of the speakerphones forward fifty times per second and fifty times backwards, meaning that we can hear the 50Hz tone on the output. Listening to a 50Hz noise is not very interesting, so you could connect to points 1 and 2 some low frequency signal source (CD player or a microphone).

There are literally thousands of different circuits using a transistor as an active, amplifying device. And all these transistors operate in a manner shown in our experiments, which means that by building this example, you're actually building a basic building block of electronics.



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